

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The glow-in-the-dark cable must remain visibly detectable from at least 2 meters away in a completely dark room for a minimum of 4 hours after 30 minutes of light exposure. This is important because it needs to be visible in areas of low light for the design to be effective. It will also address the problem directly of night time charging with an ample source of light to provide visibility to the user.

Design Requirement #2 (HIGH Priority)

The directional indicator must be easily accessible to an individual with little visibility. This is important because in conditions of low visibility, there must be a convenient way to orient a device correctly so that it can connect to a source of power without trouble of correctly placing it without vision.

Design Requirement #3 (HIGH Priority)

The device must be easily located once it is plugged in, and can be secured in an area where the individual can remember it is placed. This is important because it will allow the device to be easily accessed in a constant location without complete movement.

Rubric

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Element C: Presentation and Justification of Solution Design Requirements

C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None
C4. Objective & measurable. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
C5. Leading to solution. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	would be highly likely to lead to a tangible and	would be likely to lead to a tangible and viable	have a potential to lead to a tangible and	have some potential to lead to a tangible and	have little potential to lead to a tangible and	cannot lead to a tangible and viable problem



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	viable problem solution	problem solution	viable problem solution	viable problem solution	viable problem solution	solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.